

Team A.I. (Acoustic Innovators)

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Mission statement: To create a commercially viable Noise Canceling System (NCS) that adeptly reduces urban noise pollution that can be integrated into various modern day devices.

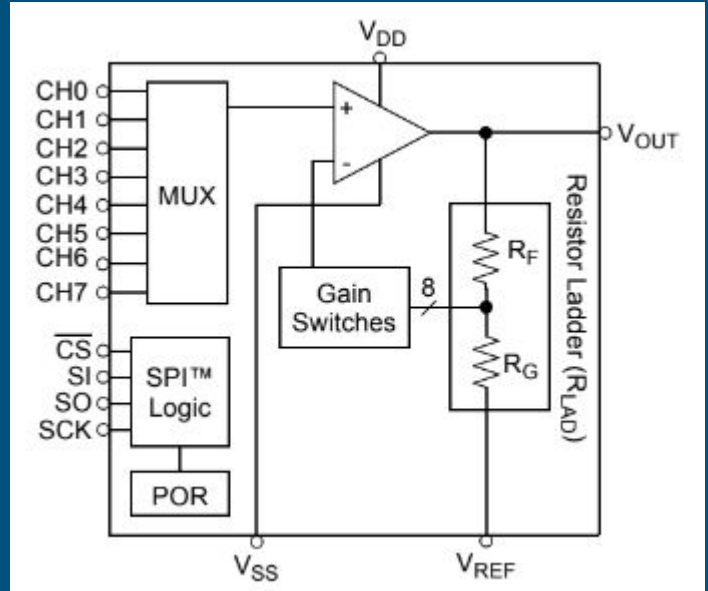
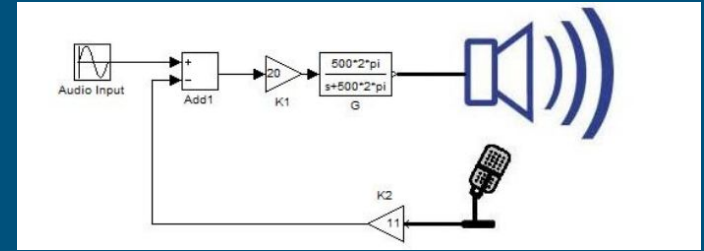
Concept

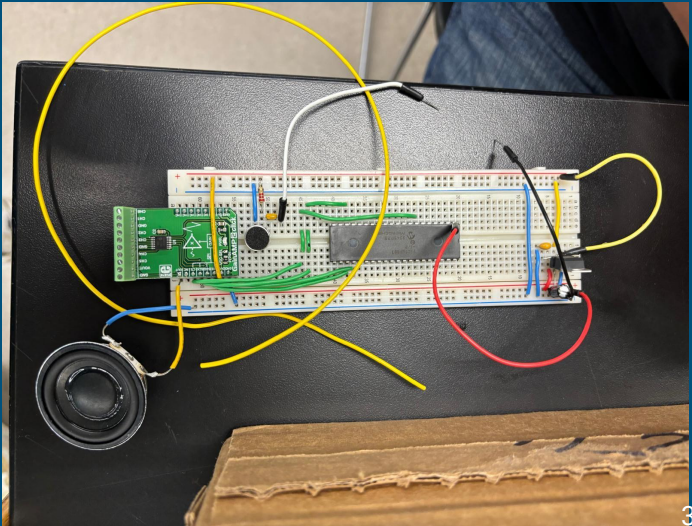
Needed features:

- Preserve the clarity of spoken words accurately.
- Minimize ambient noise levels effectively throughout the noise levels.
- Maintain high sound quality throughout the audio.

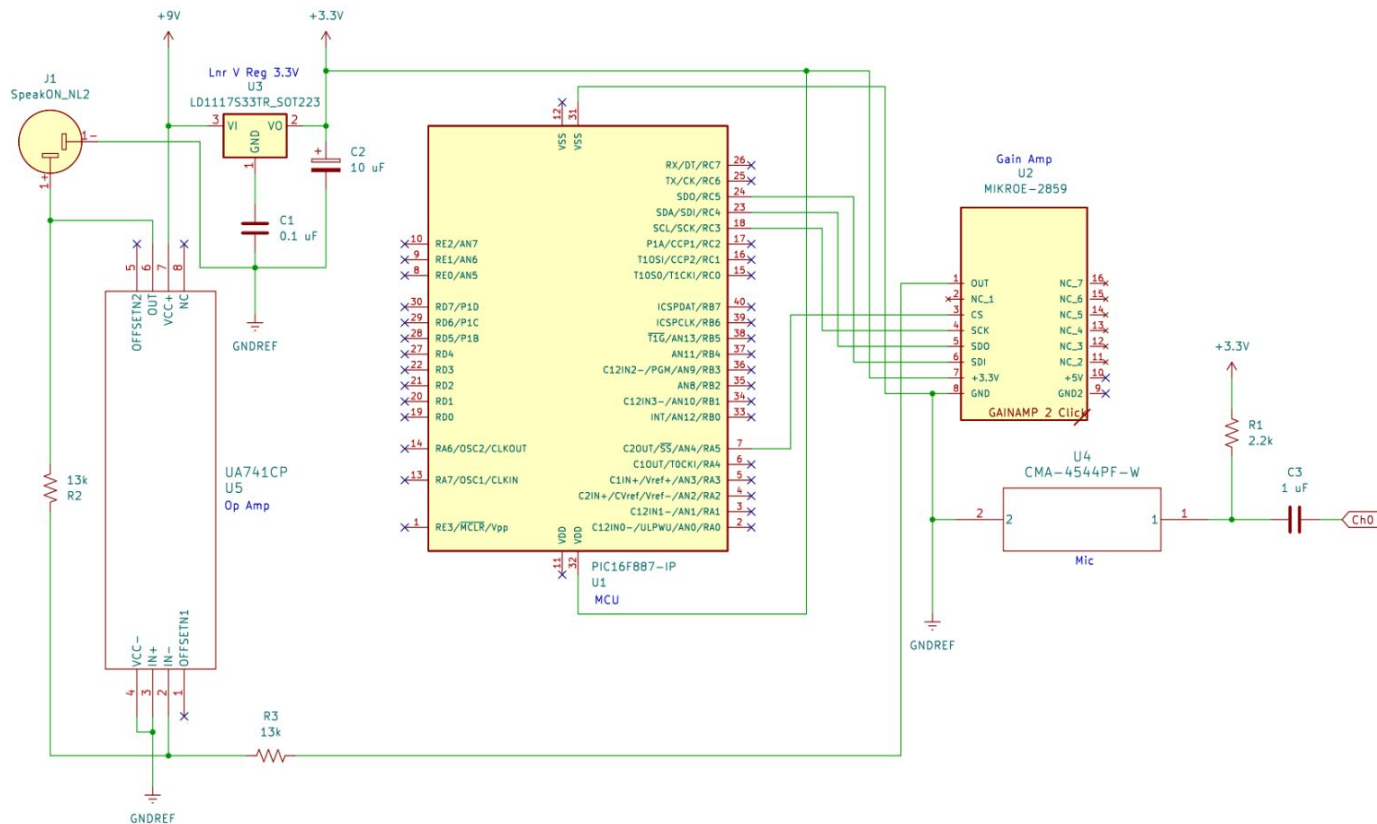
Ideal features:

- Eliminate background noise completely from the environment.
- Differentiate clearly between voice and ambient noise.
- Implement adaptive noise cancellation for dynamic environments.
- Provide user-adjustable controls for personalized settings.





Schematic



Materials for Noise Canceling System



Microcontroller, MIKROE-2859



Speaker



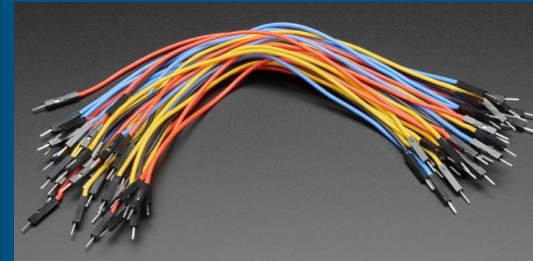
Microphone



Batteries



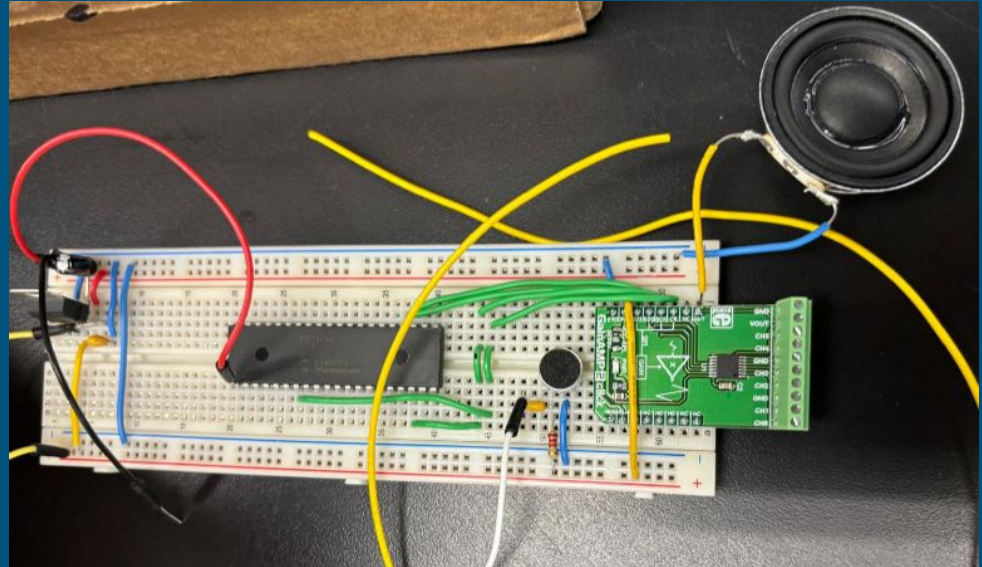
Resistors &
Capacitors



Jumper Wires

Prototype Noise Canceling System I

- The prototype works by capturing audio through the microphone, which is then processed by the microcontroller for noise cancellation. The signal, with reduced noise, is outputted through the speaker. The capacitors filter electrical noise, ensuring a clearer signal.



Setting Gain and Channel Select

Software controlled gain and channel select:

REGISTER 5-2: GAIN REGISTER

U-x	U-x	U-x	U-x	U-x	W-0	W-0	W-0
—	—	—	—	—	G2	G1	G0
bit 7					bit 0		

bit 7-3 **Unimplemented:** Read as '0' (reserved for future use)

bit 2-0 **G2-G0: Gain Select Bits**

000 = Gain of +1 (Default)
001 = Gain of +2
010 = Gain of +4
011 = Gain of +5
100 = Gain of +8
101 = Gain of +10
110 = Gain of +16
111 = Gain of +32

REGISTER 5-3: CHANNEL REGISTER

U-x	U-x	U-x	U-x	U-x	W-0	W-0	W-0
—	—	—	—	—	C2	C1	C0
bit 7					bit 0		

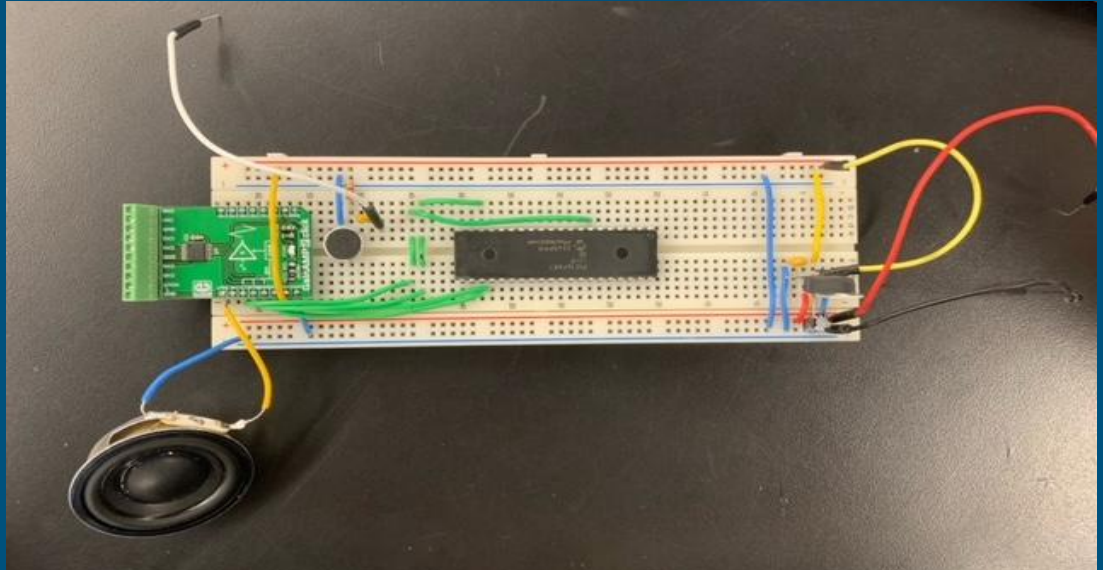
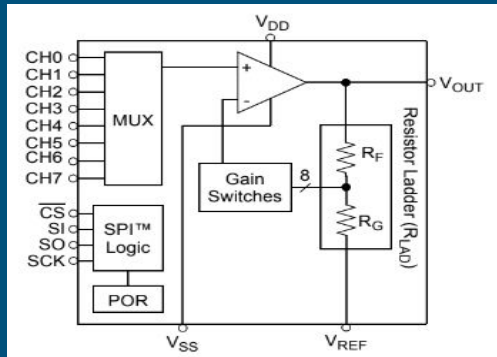
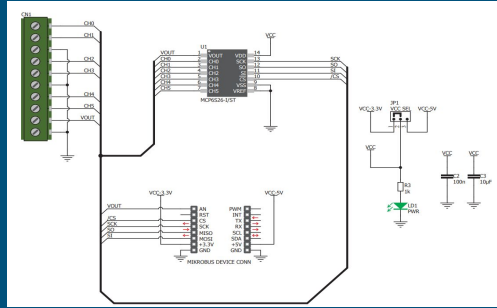
bit 7-3 **Unimplemented:** Read as '0' (reserved for future use)

bit 2-0 **C2-C0: Channel Select Bits**

MCP6S21	MCP6S22	MCP6S26	MCP6S28
000 = CH0 (Default)	CH0 (Default)	CH0 (Default)	CH0 (Default)
001 = CH0	CH1	CH1	CH1
001 = CH0	CH0	CH2	CH2
011 = CH0	CH1	CH3	CH3
100 = CH0	CH0	CH4	CH4
101 = CH0	CH1	CH5	CH5
110 = CH0	CH0	CH0	CH6
111 = CH0	CH1	CH0	CH7

GainAmp2 Chip is programmed through SPI via the PIC Microcontroller

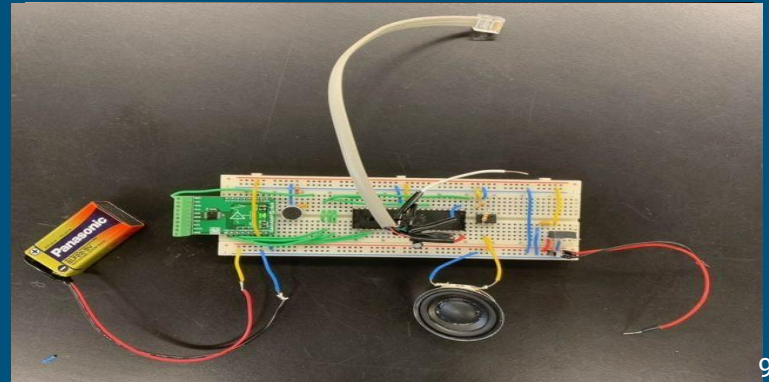
Final Noise Canceling System II



Final Noise Canceling System II Product/Improvements

Improvements Added:

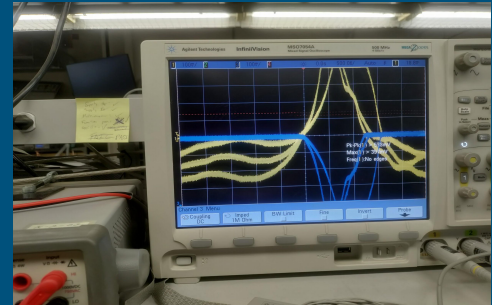
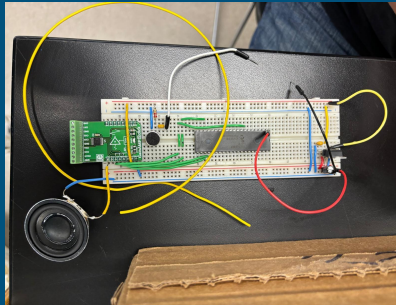
- Adding precise capacitor values for clearer signal.
- Enhanced connections for better noise accuracy.
- Use a high-fidelity speaker for better sound quality.
- Establish a stable, regulated power supply for consistency.
- Option for battery-powered operation for portability.



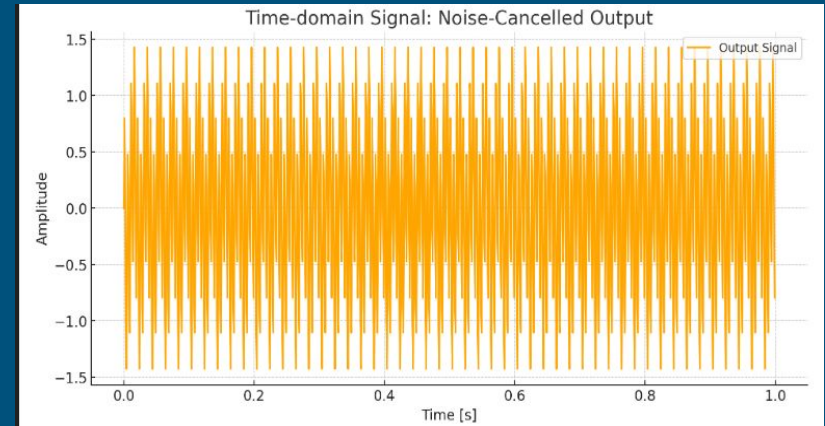
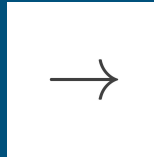
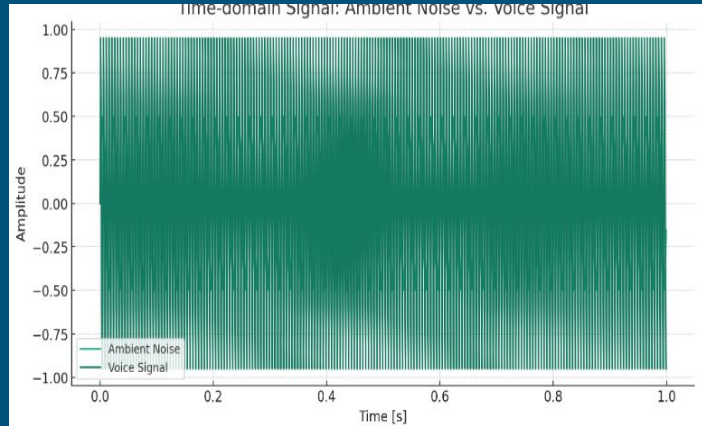
Testing Noise Canceling System

Testing Process:

- Test audio quality: Record, process, and playback sounds to evaluate clarity and noise suppression.
- Stress-test components: Run continuous cycles to ensure durability and consistent performance.
- Assess power consumption: Monitor power supply to optimize energy usage and component efficiency.



Noise Cancellation Simulation Before and After



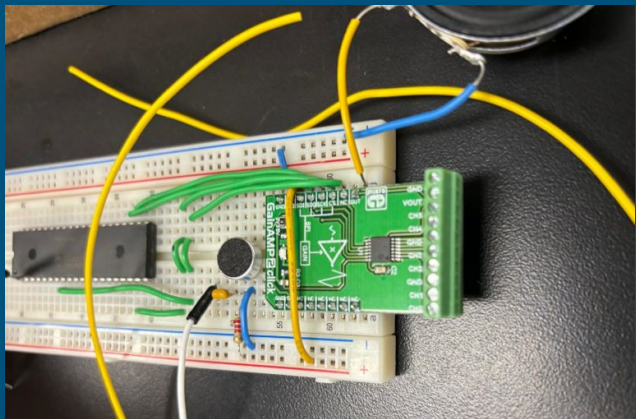
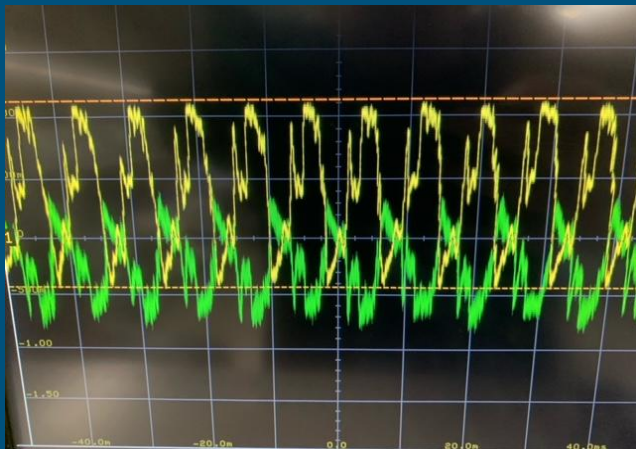
Evaluation

Functionality

- Captures sound for real-time noise filtering.
- Processes audio to preserve voice integrity.
- Outputs noise-reduced signal through a speaker.

Performance

- Measures noise reduction efficiency against benchmarks.
- Tests clarity of sound in various noise environments.



Contributions

Saif Elsaady: Constructed powerpoint, report, and general review

Scott Lidman: Circuit construction, testing, report, presentation writing, and general review

Elton Salt: Circuit construction and testing, and general review

Marla Hawthorne: Circuit construction, testing, report, presentation writing, and general review

Kevis Seymore: Circuit construction, testing, schematic creation, and general review

Angelo Navoa: prototype 2 evaluation and project improvement analysis